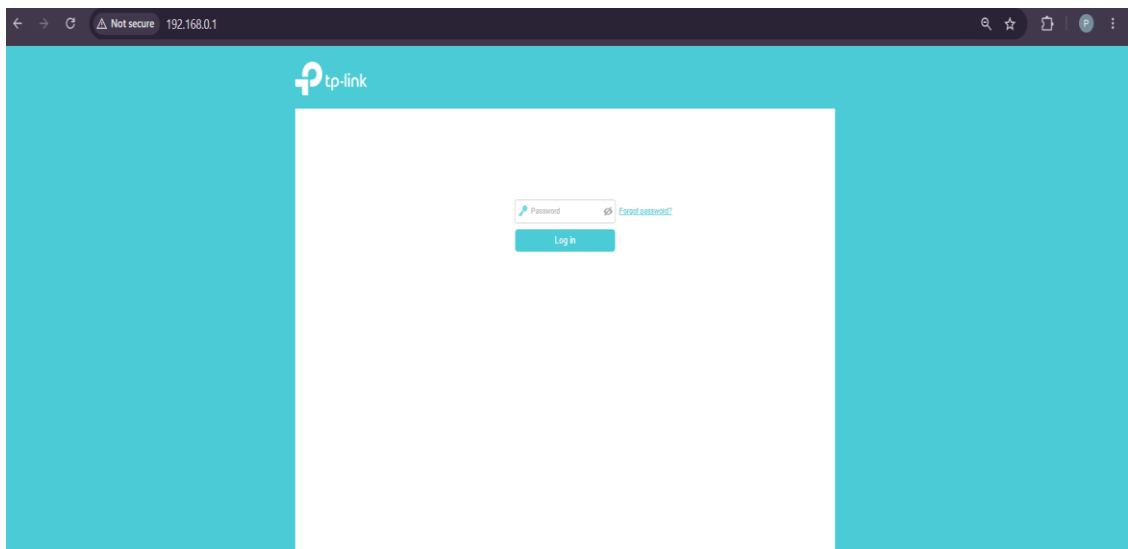


ASSIGNMENT 3

Configuring Basic Network Hardware Theory

❖ **Task (1):** Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

Ans.



❖**Task (2):** Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Ans.

- **IPv4 Address : 192.168.0.110**
- **Subnet Mask : 255.255.255.0**
- **Default Gateway : 192.168.0.1**

❖**Task (3):** Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

Ans.

- **Current IP Address:**
 - **IPv4 Address : 192.168.0.110**
 - **Subnet Mask : 255.255.255.0**
 - **Default Gateway : 192.168.0.1**

Internet Protocol Version 4 (TCP/IPv4) Properties

GeneralAlternative Configuration

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address:

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Subnet mask:

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Default gateway:

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☒ Obtain DNS server address automatically

☐ Use the following DNS server addresses:

Preferred DNS server:

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.

.

.

Alternative DNS server:

.

.

.

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Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

192 . 168 . 1 . 50

Subnet mask:

255 . 255 . 255 . 0

Default gateway:

192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

8 . 8 . 8 . 8

Alternative DNS server:

1 . 1 . 1 . 1

❖**Task (4):** In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Ans.

IP Address: 192 . 168 . 0 . 1

Subnet Mask: 255.255.255.0

IGMP Snooping: ☒ Enable

Second IP: ☐ Enable

DHCP: ☒ Enable

☒ DHCP Server ☐ DHCP Relay

IP Address Pool: 192 . 168 . 0 . 100 - 192 . 168 . 0 . 199

Address Lease Time: 120 minutes. (1-2880. The default value is 1440.)

Default Gateway: 192 . 168 . 0 . 1 (Optional)

- The router's IP address is 192.168.0.1 and the subnet mask is 255.255.255.0.
- The DHCP range is from 192.168.0.100 to 192.168.0.199.
- This means the router can automatically assign IP addresses to devices within this range.
- If two devices get the same IP address, an IP conflict can occur and the devices may have network or internet problems.